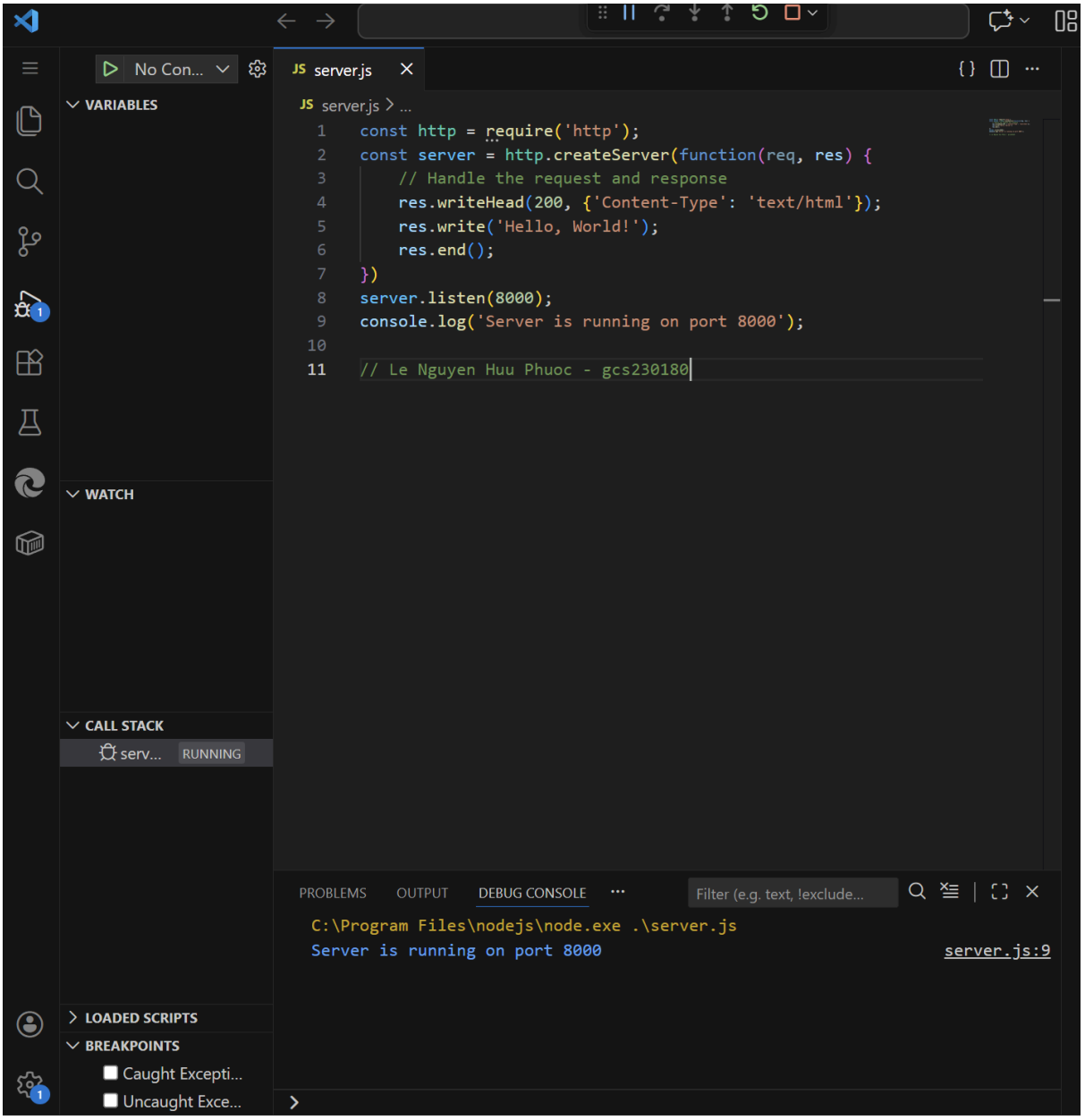
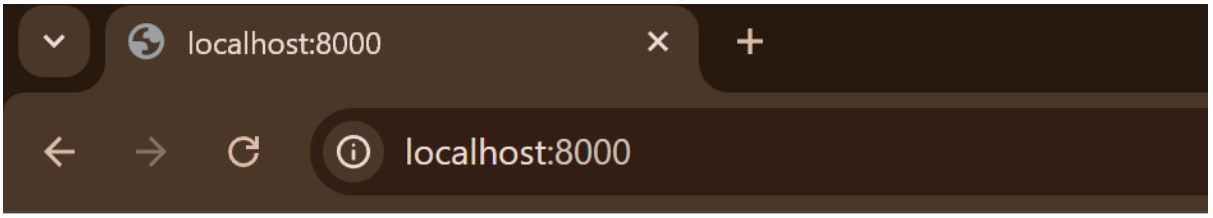
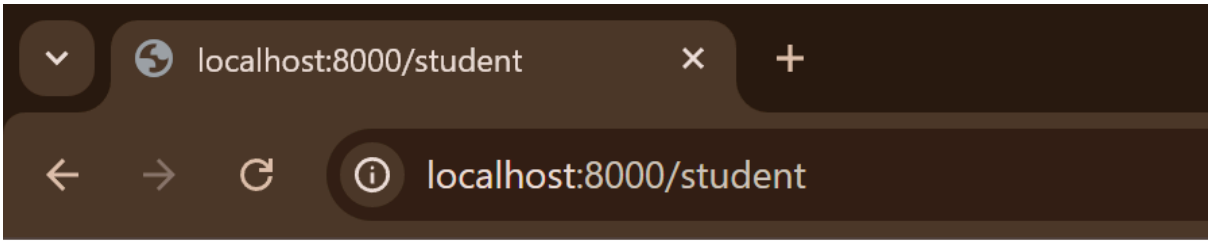
Student Name: Le Nguyen Huu Phuoc
ID: GCS230180

1.Screen Shots

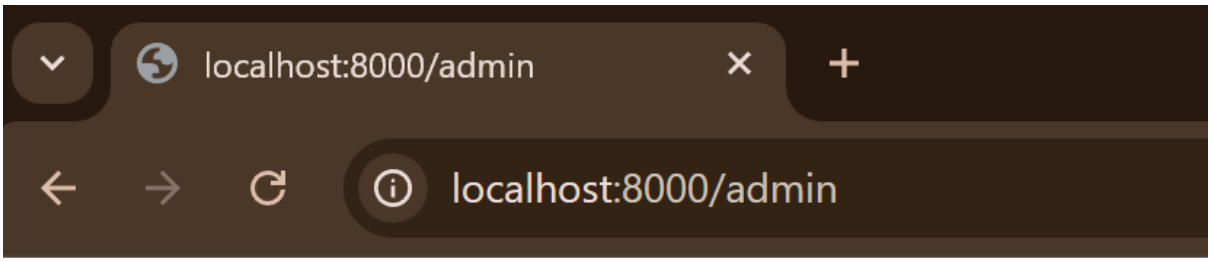




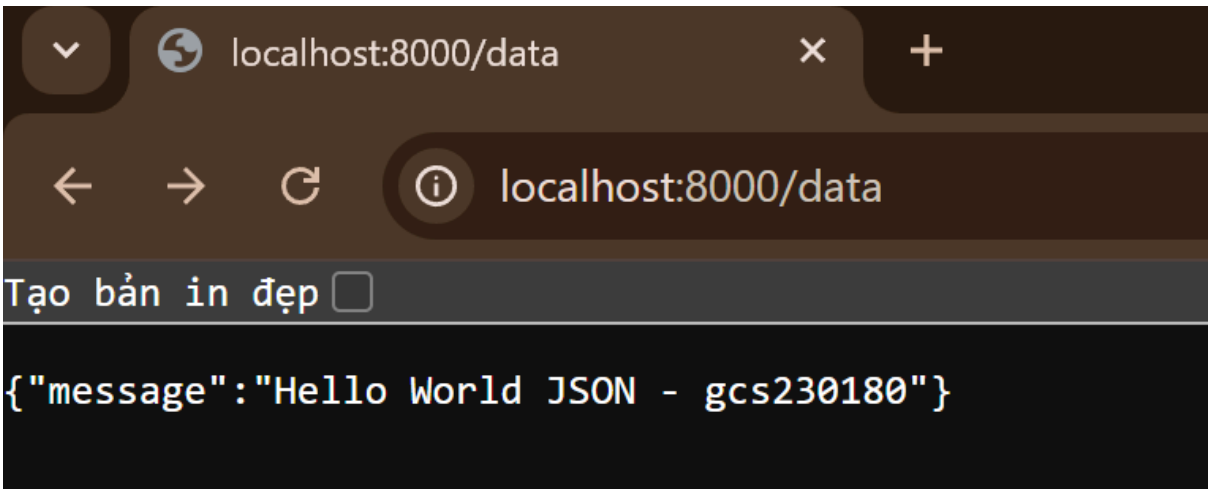
This is Home Page - gcs230180

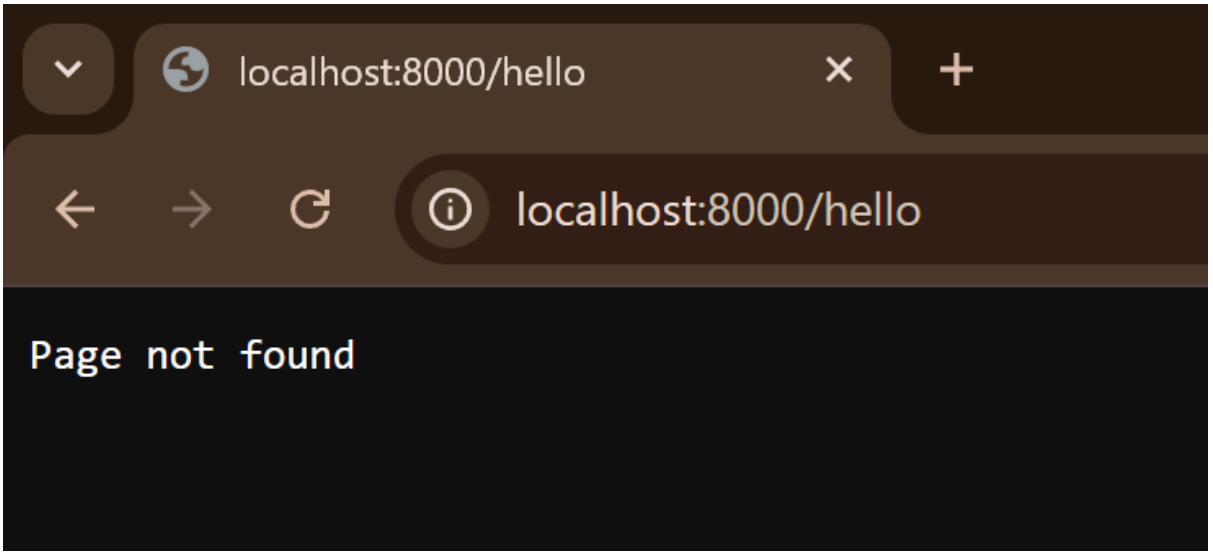


This is Student Page - gcs230180



This is Admin Page - gcs230180





2.Understanding

Through 7A and 7B, I have gained a solid foundational understanding of Node.js and how it operates as a server-side environment. I learned that Node.js is built on a single-threaded, non-blocking, and asynchronous architecture, which makes it highly efficient for handling multiple requests without consuming excessive memory.

In our practical implementation, we utilized the built-in http module to create a local web server running on port 8000. I learned how to handle incoming client requests and send appropriate responses by manipulating the response headers and body using methods like writeHead(), write(), and end().

One of the most important concepts I grasped was basic routing. By evaluating the req.url property within if and else if statements, I was able to direct the server to respond differently based on the URL path the user visited. This allowed me to create distinct outputs for the Home, Student, and Admin pages. Furthermore, I successfully configured the server to output structured JSON data when accessing the /data route, and implemented error handling to return an "Invalid Request!" message for unrecognized URLs. Overall, this task provided me with valuable hands-on experience in building the core logic of a back-end server.